1 Research project's background

This project studies how external uncertainty shocks originating in the United States and China are transmitted to Taiwan, which external sources matter most for Taiwan's domestic uncertainty, and whether the dominant transmission channel changes over time. These questions are especially important for Taiwan because it is deeply integrated with both the U.S. and Chinese economies. As a result, Taiwan lies directly in the path of major policy and geopolitical developments between the two countries. When the U.S. Federal Reserve tightens monetary policy, when trade disputes between the U.S. and China escalate, or when cross-strait political tensions rise, Taiwan often faces immediate consequences for its trade, investment, and financial flows.

However, major policy shifts and geopolitical realignments rarely move from initial proposal to a well-defined policy trajectory overnight. Central banks move in sequences of decisions, trade conflicts unfold through rounds of negotiation and retaliation, and cross-strait relations evolve over extended periods rather than at a single point in time. As a result, there is often a long interval between the announcement or anticipation of a policy change and the point at which a stable policy regime actually emerges. Along the way, planned policy changes may be revised, delayed, or even cancelled altogether. For example, rounds of U.S.–China tariff threats have repeatedly ended with smaller or postponed tariff increases than initially announced, and cross-strait initiatives have at times been shelved after domestic political opposition. During this interval, firms, households, and investors face heightened uncertainty about the future path of policy, demand, and financial conditions. This heightened uncertainty, in turn, affects their investment, consumption, and portfolio decisions (Bloom, 2009, 2014). One prominent example is trade policy uncertainty, which strongly influences firms' trade and investment decisions (Handley and Limao, 2022). To capture these evolving dynamics without sacrificing the macroeconomic dimension, this project utilizes a monthly dataset. Monthly data are granular enough to capture short-lived spikes in uncertainty during negotiation periods that quarterly data often smooth out, while still allowing the analysis to include key real variables such as industrial production and trade balances that are generally unavailable at daily frequency.

Empirically, however, identifying external uncertainty shocks and quantifying their effects on

Taiwan presents several methodological challenges. First, external and domestic sources of uncertainty are tightly intertwined, creating a severe challenge for identification. A single geopolitical event, such as a shift in U.S.–China relations, often simultaneously triggers global financial volatility and shifts in Taiwan's domestic political landscape. In a small-scale model that lacks sufficient domestic controls, the economic impact of these internal political shifts would be erroneously attributed solely to the external shock. This form of omitted variable bias can lead to a misdiagnosis of the true drivers of uncertainty, potentially overstating the direct influence of external factors while neglecting local transmission mechanisms (Carriero, Clark and Marcellino, 2018). Addressing this issue requires expanding the information set to include a rich array of both domestic and external variables, necessitating the use of a large-scale econometric model.

Second, while a large-scale model is necessary to address this bias, it introduces identification trade-offs. In large systems, standard identification methods based on Cholesky decomposition become problematic because the results are sensitive to variable ordering, creating a conceptual flaw known as order dependence. Addressing this issue requires an order-invariant framework. However, order invariance is mathematically incompatible with the strict block-exogeneity restrictions commonly used in small open economy models to prevent domestic variables from affecting global ones. Consequently, instead of imposing these rigid constraints, we employ the data-driven identification strategy proposed by Davidson, Hou and Koop (2025), treating external drivers as "unclassified variables". This avoids arbitrary restrictions on the contemporaneous relationships between variables, allowing for a data-determined structure that can accommodate potential feedback loops, rather than ruling them out by assumption. This approach suits Taiwan's pivotal role in global technology supply chains, where strict zero restrictions might assume away significant feedback effects originating from the supply side.

Third, many existing approaches face two related limitations. One is their reliance on proxy measures of uncertainty. The other is their use of a two-step procedure that first estimates uncertainty and then evaluates its macroeconomic effects in a separate model. Regarding the former limitation, widely used text-based indices built from international news may not reflect domestic conditions. For example, during Nancy Pelosi's 2022 visit, international coverage emphasized imminent war risk, while sentiment in Taiwan remained relatively calm. Such reliance on external proxies can create a "perception gap" and distort estimates of uncertainty's impact on the local economy. Regarding the latter limitation, the two-step design treats the estimated uncertainty series as observed data, ignores estimation uncertainty, and can induce measurement-error bias and model inconsistency (Carriero, Clark and Marcellino, 2018). To address these issues, we employ the proposed stochastic volatility in mean vector autoregression (SVMVAR) framework to jointly estimate uncertainty and its economic effects in a single step. By modeling uncertainty as a latent factor driven by the data itself, we avoid reliance on potentially misaligned external proxies, ensuring that our measure reflects actual domestic economic and financial conditions.

Finally, existing uncertainty measures, such as global financial volatility indices or country-specific policy uncertainty indices, do not distinguish between the *channels* through which uncertainty transmits to the economy. Widely adopted indices such as the VIX or the Baker, Bloom and Davis (2016) Economic Policy Uncertainty index compress the multidimensional nature of uncertainty into a single scalar, obscuring whether a given shock propagates through real activity or financial markets. For a small open economy like Taiwan, this distinction carries different policy implications. When external shocks operate primarily through **macroeconomic channels**, affecting trade flows, export demand, and production linkages, the appropriate policy response involves instruments oriented toward the real economy, such as export facilitation and structural adjustment support. When shocks instead propagate through **financial channels**, disrupting capital flows, asset prices, and credit conditions, the Central Bank of Taiwan faces a different set of imperatives, including foreign exchange intervention and liquidity management. Existing empirical frameworks, however, are ill-equipped to make this distinction in a time-varying setting, leaving policymakers without a systematic basis for identifying which channel is dominant at any given point in time. This constitutes the fourth and most substantive methodological gap that the present project is designed to fill.

To address these challenges, this project applies the order-invariant stochastic volatility in mean vector autoregression (OI-SVMVAR) framework developed by Davidson, Hou and Koop (2025) to the Taiwan context. This project treats external driverssuch as U.S. monetary policy indicators and cross-strait tension indicesas unclassified variables and uses their time-varying co-movement with Taiwans macroeconomic and financial common factors to identify the operative transmission channel at each point in time without requiring the researcher to impose it ex ante.

While the OI-SVMVAR identifies the presence and macroeconomic impact of external uncertainty shocks, it is silent on the underlying micro-founded frictions driving these responses. Conventional approaches attempt to bridge this empirical-to-theoretical gap by matching impulse response functions (IRFs) of standard variables with those from dynamic stochastic general equilibrium (DSGE) models. However, variable-level IRF matching is often confounded by general equilibrium effects, making it difficult to isolate the exact structural friction.

To bridge this gap, this project develops a novel, theorem-based framework: the **Wedge Projection Theorem**. We formalize the exact mathematical mapping between a nonlinear structural model with stochastic volatility and a first-order Business Cycle Accounting (BCA) prototype. We prove that when a specific friction (e.g., a collateral constraint versus a working capital requirement) generates second-order risk corrections in the true economy, these corrections project mathematically onto specific first-order wedges (e.g., the investment wedge versus the labor wedge). This breakthrough allows us to derive falsifiable "wedge signatures" to pinpoint exactly where and how external uncertainty distorts the domestic economy. Furthermore, by projecting the empirically extracted uncertainty factors from the VAR onto the historical BCA wedges using full Markov Chain Monte Carlo (MCMC) uncertainty propagation, we overcome the generated regressor bias typical of two-step estimation strategies.

This project makes four contributions. First, the large-scale specification mitigates the omitted-variable bias that is especially severe when external and domestic sources of uncertainty are intertwined (Carriero, Clark and Marcellino, 2018). Second, the order-invariant identification strategy avoids the distortions associated with variable ordering in large VAR systems. Third, the model's time-varying classification results help identify whether policymakers in Taiwan should respond primarily with real-economy instruments or financial stability tools. Fourth, the Wedge Projection Theorem provides theoretically grounded, testable wedge signatures to rigorously disentangle competing transmission channels of uncertainty, offering a unified method to validate micro-founded frictions against empirical data.

The remainder of this proposal details the three-year research design. The **first year** focuses on empirical extraction: assembling a monthly dataset of over forty domestic and global variables, and executing the OI-SVMVAR to extract Taiwan's time-varying uncertainty factors ($h_{m,t}$, $h_{f,t}$) and classification probabilities. The **second year** focuses on theoretical derivation: solving nonlinear DSGE models via third-order perturbation to formally prove the Wedge Projection Theorem and derive the mutually exclusive wedge signatures for competing financial frictions. The **third year** is dedicated to empirical validation: constructing the BCA prototype for Taiwan, executing the MCMC-integrated local projections of the Year 1 uncertainty factors onto the empirical wedges, and conclusively testing the data against the Year 2 theoretical signatures to identify the dominant structural transmission channel.

The First Year

1.1 Methods, procedures, and implementation schedule

(1) Research principles, methods, and the innovation of research methods

(1.1) The OI-SVMVAR Framework

To address the identification challenges outlined in Section 1, this project employs the **order-invariant stochastic volatility in mean vector autoregression** (OI-SVMVAR) framework developed by Davidson, Hou and Koop (2025). We select this framework because it uniquely resolves the three methodological obstacles that confront any attempt to trace external uncertainty shocks through a small open economy: it accommodates a large information set, eliminating the omitted variable bias that confounds small-scale models; its order-invariant identification ensures that results do not depend on arbitrary variable ordering; and its time-varying classification mechanism allows the data—rather than the researcher—to determine whether a given external shock transmits through macroeconomic or financial channels. The model consists of five coupled components, estimated jointly in a single Bayesian step, which we now present in turn.

To capture the direct impact of uncertainty on real economic activity and financial conditions, the observation equation includes contemporaneous and lagged latent uncertainty factors in the conditional mean of a VAR, following the specification of Davidson, Hou and Koop (2025):

$$\mathbf{y}_t = \sum_{l=1}^{p} \mathbf{B}_l \, \mathbf{y}_{t-l} + \sum_{j=0}^{q} \mathbf{A}_j^h \, \mathbf{h}_{t-j} + \mathbf{B}_0^{-1} \boldsymbol{\varepsilon}_t^y, \qquad \boldsymbol{\varepsilon}_t^y \sim \mathcal{N}(\mathbf{0}, \, \mathbf{U}_t), \tag{1}$$

where $\mathbf{y}_t$ is the $N \times 1$ vector of observables. In our application, $N = 47$, covering Taiwans domestic economy, U.S. and Chinese indicators, and global risk measures. The matrices $\mathbf{B}_l$ are $N \times N$ VAR coefficient matrices for lag $l = 1, \ldots, p$, and $\mathbf{B}_0$ is the contemporaneous impact matrix. Crucially, $\mathbf{B}_0$ is specified as a full, unrestricted nonsingular matrix—subject only to the normalization that its diagonal elements equal one—rather than as a lower triangular Cholesky matrix. This unrestricted specification delivers **order invariance**: because the model imposes no Cholesky ordering on $\mathbf{B}_0$, inference on uncertainty, impulse responses, variance decompositions, and classification probabilities does not depend on the arbitrary position of a variable within $\mathbf{y}_t$.

The vector $\mathbf{h}_t = (h_{m,t}, \, h_{f,t})'$ is a 2×1 vector containing the two latent common log-volatility factors that represent Taiwan's **macroeconomic uncertainty** ($h_{m,t}$) and **financial uncertainty** ($h_{f,t}$), respectively. Following Davidson, Hou and Koop (2025), Equation (1) excludes an intercept. We standardize the transformed observables to have zero mean before estimation. Thus, the conditional mean depends on lagged observables and common log-volatilities, including both their contemporaneous and lagged values.

The $N \times 2$ matrices $\mathbf{A}_0^h, \ldots, \mathbf{A}_q^h$ capture the **stochastic-volatility-in-mean** effect. Because contemporaneous and lagged common log-volatilities enter the conditional mean of $\mathbf{y}_t$, higher uncertainty can directly shift the expected path of all observables, including output, employment, asset prices, and credit conditions. This feature distinguishes the SVMVAR from a standard stochastic volatility model, where volatility enters only the error covariance and therefore cannot directly affect the levels of economic variables.

The term $\sum_{j=0}^{q} \mathbf{A}_j^h \mathbf{h}_{t-j}$ allows us to test a key prediction from real-options and precautionary-savings theories (Bloom, 2009, 2014): heightened uncertainty can depress investment, consumption, and hiring not only by increasing dispersion but also by shifting the conditional mean of economic outcomes. In the baseline specification, we follow Davidson, Hou and Koop (2025) and set $p = 6$

and $q = 2$. We also conduct robustness checks with shorter lag lengths, given the shorter Taiwan sample.

The diagonal variance matrix $\mathbf{U}_t$ is partitioned conformably with the macroeconomic, unclassified, and financial blocks:

$$\mathbf{U}_t = \begin{pmatrix} \mathbf{\Omega}_{m,t} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{\Omega}_{u,t} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Omega}_{f,t} \end{pmatrix}, \tag{2}$$

where the variance blocks are diagonal matrices,

$$\mathbf{\Omega}_{m,t} = \operatorname{diag}\left(e^{\omega^m_{1,t}}, \ldots, e^{\omega^m_{n_m,t}}\right), \qquad \mathbf{\Omega}_{u,t} = \operatorname{diag}\left(e^{\omega^u_{1,t}}, \ldots, e^{\omega^u_{n_u,t}}\right),$$

$$\mathbf{\Omega}_{f,t} = \operatorname{diag}\left(e^{\omega^f_{1,t}}, \ldots, e^{\omega^f_{n_f,t}}\right).$$

Thus, as in Davidson, Hou and Koop (2025), $\omega^m_{i,t}$, $\omega^u_{i,t}$, and $\omega^f_{i,t}$ denote the log-volatilities of the macroeconomic, unclassified, and financial variables, respectively, while $e^{\omega^k_{i,t}}$ is the corresponding conditional variance. The block ordering in Equation (2) follows the convention $\mathbf{y}_t = (\mathbf{y}'_{m,t}, \mathbf{y}'_{u,t}, \mathbf{y}'_{f,t})'$, but this is a notational ordering rather than an identifying Cholesky ordering. Because $\mathbf{B}_0$ remains unrestricted, the block structure in $\mathbf{U}_t$ classifies volatilities but does not impose an ordering on contemporaneous interactions. The implications of this design choice are discussed in detail in subsection (1.3).

The structural heart of the model lies in the specification linking each variable's log-volatility to the common uncertainty factors. Following Davidson, Hou and Koop (2025), the N variables in $\mathbf{y}_t$ are partitioned into three blocks—n_m macroeconomic variables, n_f financial variables, and n_u unclassified variables, with $n_m + n_f + n_u = N$—and the log conditional variance of variable i is decomposed into a **variable-specific idiosyncratic component** $\eta_{i,t}$ and a loading on the relevant common factor:

$$\begin{aligned} \omega^m_{i,t} &= \eta^m_{i,t} + h_{m,t}, \quad i = 1, \ldots, n_m, \\ \omega^f_{i,t} &= \eta^f_{i,t} + h_{f,t}, \quad i = 1, \ldots, n_f, \\ \omega^u_{i,t} &= \eta^u_{i,t} + h_{s_{i,t},t}, \quad i = 1, \ldots, n_u. \end{aligned} \tag{3}$$

The idiosyncratic terms $\eta^k_{i,t}$ ($k \in \{m, f, u\}$) capture variable-specific volatility movements that are orthogonal to the common factors, ensuring that the log-volatilities of variables within the same block are not forced into perfect collinearity. As in Davidson, Hou and Koop (2025), these equations can be viewed as a factor-volatility model whose common-factor loadings are normalized to one; this normalization preserves the interpretation of $h_{m,t}$ and $h_{f,t}$ as common log-volatility components and avoids an additional weak-identification problem. Each $\eta^k_{i,t}$ follows a stationary AR(1) process, so that idiosyncratic volatility shocks are mean-reverting while the common factors $h_{m,t}$ and $h_{f,t}$ can exhibit the persistent swings characteristic of aggregate uncertainty episodes. Variables pre-classified as macroeconomic load exclusively on $h_{m,t}$, so that their common volatility component rises and falls only with aggregate real-economy uncertainty. Variables pre-classified as financial load exclusively on $h_{f,t}$. For unclassified variables, however, the common factor is selected by a discrete latent state indicator $s_{i,t} \in \{m, f\}$: when $s_{i,t} = m$, the variable's common volatility component is $h_{m,t}$; when $s_{i,t} = f$, it is $h_{f,t}$. This discrete switching mechanism—rather than a continuous blend of both factors—means that at each point in time the model assigns each unclassified variable entirely to one block, and the *posterior probability* of that assignment becomes the object of inferential interest. As we explain in subsection (1.2), this is the mechanism through which we identify whether a given external uncertainty source transmits to Taiwan through the macroeconomic channel or the financial channel.

The two common log-volatility factors evolve jointly according to the dynamic process used by Davidson, Hou and Koop (2025):

$$\mathbf{h}_t = \sum_{r=1}^{p_h} \mathbf{\Phi}_r \mathbf{h}_{t-r} + \sum_{s=1}^{p_y} \mathbf{\Psi}_s \mathbf{y}_{t-s} + \boldsymbol{\varepsilon}^h_t, \qquad \boldsymbol{\varepsilon}^h_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_h), \tag{4}$$

where $\mathbf{\Phi}_r$ allows macroeconomic and financial uncertainty to affect each other over time, $\mathbf{\Psi}_s$ allows past values of the observed VAR variables to predict the common log-volatility factors, and $\mathbf{\Sigma}_h$ permits shocks to the two uncertainty factors to be correlated. This specification is important for two reasons. First, it does not force uncertainty to be an exogenous random walk: past macroeconomic, financial, and external conditions may help forecast Taiwan's domestic uncertainty, which is essential in a small open economy. Second, it preserves the SVMVAR's joint-estimation logic:

the common uncertainty factors and their effects on $\mathbf{y}_t$ are estimated simultaneously rather than constructed in a first step and inserted into a second-stage VAR. In the baseline implementation we follow Davidson, Hou and Koop (2025) by setting $p_h = 2$ and $p_y = 1$. For impulse response analysis, the reduced-form innovation $\boldsymbol{\varepsilon}_t^h$ is decomposed as $\boldsymbol{\varepsilon}_t^h = \mathbf{L}\mathbf{e}_t^h$, where $\mathbf{L}\mathbf{L}' = \boldsymbol{\Sigma}_h$ and $\mathbf{e}_t^h = (e_{m,t}^h, e_{f,t}^h)'$ are orthogonal macroeconomic and financial uncertainty shocks. The baseline ordering follows Davidson, Hou and Koop (2025): macroeconomic uncertainty can affect financial uncertainty contemporaneously, while financial uncertainty affects macroeconomic uncertainty with a lag. Because $h_{m,t}$ and $h_{f,t}$ are defined in logs, the implied level of uncertainty $\exp(\frac{1}{2}h_{k,t})$ is guaranteed to be positive at all times.

The final component governs the time-varying classification of unclassified variables, which is the mechanism that transforms the statistical model into an economic identification device. For each unclassified variable i, the latent state indicator $s_{i,t} \in \{\text{macro, financial}\}$ in Equation (3) follows a first-order two-state **Markov chain** with transition probabilities $q_{mm}^{(i)} = P(s_{i,t} = m \mid s_{i,t-1} = m)$ and $q_{ff}^{(i)} = P(s_{i,t} = f \mid s_{i,t-1} = f)$. The posterior classification probability

$$\pi_{i,t} = P(s_{i,t} = \text{macro} \mid \mathbf{y}_{1:T}) \tag{5}$$

is the posterior probability—conditional on the full observed sample $\mathbf{y}_{1:T}$—that unclassified variable i belongs to the macroeconomic block at time t. In the MCMC algorithm, the full path $(s_{i,1}, \dots, s_{i,T})$ and its transition probabilities are drawn using the Markov-switching sampler of Chib (1996), as in Davidson, Hou and Koop (2025). This specification implies that a variable may be classified as macroeconomic during one episode and as financial during another, with the timing and persistence of regime switches determined entirely by the data. The transition probabilities $q_{mm}^{(i)}$ and $q_{ff}^{(i)}$ are estimated jointly with all other model parameters within the Bayesian MCMC framework, so that the persistence of each variable's classification is itself a model output rather than a researcher-imposed restriction. In our application, the time path of $\pi_{i,t}$ for each external variable constitutes the primary object of interest: it reveals, at each point in the sample, whether the uncertainty associated with a given external driver transmits to Taiwan predominantly through the macroeconomic channel ($\pi_{i,t} \approx 1$) or through the financial channel ($\pi_{i,t} \approx 0$).

The full model—comprising Equations (1)–(5)—is estimated jointly in a single-step Bayesian Markov chain Monte Carlo (MCMC) procedure developed by Davidson, Hou and Koop (2025). The algorithm builds on the computationally efficient precision-based sampler of Cross et al. (2023), which exploits band and sparse matrix structures to draw the common log-volatilities in a single block, making estimation of very large SVMVARs feasible without resorting to dimensionality reduction techniques that would obscure the specific structural shocks we aim to identify. Davidson, Hou and Koop (2025) extend this approach with a novel parameter transformation for sampling the unrestricted $\mathbf{B}_0$ and with the Markov-switching classification mechanism described above. We do not derive the prior distributions or the posterior sampling steps here; the complete Bayesian implementation is detailed in the online appendix of Davidson, Hou and Koop (2025). All subsequent analysis in this project—time-varying classification probabilities, forecast error variance decompositions, and impulse response functions—is conducted on draws from the joint posterior distribution, ensuring that parameter uncertainty and classification uncertainty are fully propagated through to the final inferential objects.

(1.2) Variable Classification and the Key Innovation

The weight that the OI-SVMVAR places on its classification trichotomy—macroeconomic, financial, and unclassified—demands that the assignment of variables to the three blocks be dictated by the economic question at hand rather than by statistical convenience. Equation (3) imposes a clean mapping between blocks and the two common log-volatility factors: pre-classified macroeconomic variables load exclusively on $h_{m,t}$; pre-classified financial variables load exclusively on $h_{f,t}$; and unclassified variables load on a time-varying mixture of the two, governed by the posterior probability $\pi_{i,t}$ defined in Equation (5). Our dataset comprises $N = 47$ monthly series partitioned into $n_m = 17$ macroeconomic variables, $n_f = 8$ financial variables, and $n_u = 22$ unclassified variables. The full list of series, together with their assigned block, is reported in Table 1.

Table 1: Variable names and data sources: 47 monthly series, 2003M1–2025M12.

Variable name	**Data source**
Panel A. Macroeconomic variables (17)	
Industrial Production Index	DGBAS, Industry and Service Statistics
Manufacturing Production Index	DGBAS, Industry and Service Statistics
Export Orders Index	Ministry of Economic Affairs (MOEA), Department of Statistics
Retail Sales (real)	DGBAS, Industry and Service Statistics
Food and Beverage Services Sales (real)	DGBAS, Industry and Service Statistics
Exports (customs, real USD)	Ministry of Finance (MOF), Customs Administration
Imports (customs, real USD)	Ministry of Finance (MOF), Customs Administration
Business Cycle Indicator	National Development Council (NDC)
CPI (YoY)	DGBAS, Price Statistics
Core CPI (YoY)	DGBAS, Price Statistics
WPI (YoY)	DGBAS, Price Statistics
Import Price Index (YoY)	DGBAS, Price Statistics
Export Price Index (YoY)	DGBAS, Price Statistics
Unemployment Rate (SA)	DGBAS, Manpower Survey Statistics
Manufacturing Employment (SA)	DGBAS, Manpower Survey Statistics
Services Employment (SA)	DGBAS, Manpower Survey Statistics
Real Manufacturing Wages (YoY)	DGBAS, Earnings and Productivity Statistics
Panel B. Financial variables (8)	
Overnight Interbank Rate	Central Bank of the Republic of China (Taiwan), Financial Statistics

Table 1 continued.

Variable name	**Data source**
10-Year Government Bond Yield	Central Bank of the Republic of China (Taiwan), Financial Statistics
Term Spread (10-year government bond yield minus overnight rate)	Authors' calculation from CBC series
TAIEX Monthly Return	TWSE total return index (MFI94U)
TAIEX Monthly Realized Volatility	Authors' calculation from daily TWSE total return index (MFI94U)
TAIEX Average Daily Turnover	TWSE market turnover statistics (FMTQIK)
Foreign Institutional Net Buying (NTD bn)	Taiwan Stock Exchange (TWSE)
Margin Buying Balance (YoY)	Taiwan Stock Exchange (TWSE)
Panel C. Unclassified variables (22)	
Taiwan domestic ambiguous variables	
CBC Discount Rate	Central Bank of the Republic of China (Taiwan), Financial Statistics
M1B Money Supply (YoY)	Central Bank of the Republic of China (Taiwan), Financial Statistics
M2 Money Supply (YoY)	Central Bank of the Republic of China (Taiwan), Financial Statistics
Bank Loans and Investments (YoY)	Central Bank of the Republic of China (Taiwan), Financial Statistics
Consumer Loans (YoY)	Central Bank of the Republic of China (Taiwan), Financial Statistics
TAIEX Level (log)	TWSE price index (MI_5MINS_HIST)
Housing Price Index (YoY)	Sinyi Realty housing price index, interpolated from quarterly to monthly frequency

Table 1 continued.

Variable name	**Data source**
Taiwan exchange-rate and market-funding variables	
NTD/USD Exchange Rate (nominal)	FRED and Central Bank of the Republic of China (Taiwan)
NTD/USD Realized Volatility	Authors' calculation from daily NTD/USD exchange rates, FRED/CBC
Real Effective Exchange Rate	Bank for International Settlements (BIS), effective exchange-rate statistics
Short Selling Balance (YoY)	Taiwan Stock Exchange (TWSE)
U.S. block	
Federal Funds Rate (effective)	FRED, Federal Reserve Bank of St. Louis (FEDFUNDS)
U.S. Industrial Production (YoY)	FRED, Federal Reserve Bank of St. Louis (INDPRO)
U.S. Credit Spread	FRED, Federal Reserve Bank of St. Louis (BAA10YM)
U.S. Economic Policy Uncertainty	*PolicyUncertainty.com*, Baker–Bloom–Davis EPU index
China block	
China Industrial Production (YoY)	National Bureau of Statistics of China (NBS) and FRED
China Producer Price Index (YoY)	National Bureau of Statistics of China (NBS) and CEIC
China M2 Money Supply (YoY)	FRED, Federal Reserve Bank of St. Louis, and People's Bank of China (PBoC)
Global indicators	
VIX (monthly average)	FRED, Federal Reserve Bank of St. Louis (VIXCLS)

Table 1 continued.

Variable name	**Data source**
Global Geopolitical Risk	Caldara and Iacoviello (2022), Geopolitical Risk dataset
Global Economic Policy Uncertainty	*PolicyUncertainty.com*, Global EPU index
Risk	
U.S. Trade Policy Uncertainty	Baker–Bloom–Davis categorical EPU series, trade-policy category

Crucially, every external driver—whether a U.S. variable, a Chinese variable, a global risk indicator, or a trade-policy measure—is assigned to the unclassified block, while Taiwan's clearly real-sector and clearly financial-market variables anchor the two classified blocks. This allocation is not a stylistic choice but the empirical core of the research design. The transmission channel of external uncertainty shocks—whether they propagate to Taiwan through real economic activity or through financial markets—is precisely the object that this project sets out to identify. Pre-assigning the U.S. Federal Funds Rate to the financial block, or China's Industrial Production Index to the macroeconomic block, would hardwire the answer into the prior structure of the model: the very question of whether U.S. monetary tightening reaches Taiwan through export demand or through exchange-rate and credit-market pressure would be foreclosed by assumption. By placing these variables in the unclassified block instead, the model allows the Markov-switching probability $\pi_{i,t}$ in Equation (5) to determine the operative transmission channel endogenously, at each point in time, based on how each external series co-moves with Taiwan's two common log-volatility factors in the data. A small set of Taiwan-domestic variables whose economic nature is intrinsically ambiguous—policy rates, monetary aggregates, credit aggregates, equity levels, house prices, exchange rates, and short-selling balances—is also placed in the unclassified block, following the spirit of Davidson, Hou and Koop (2025). This prevents the empirical anchors from being contaminated by variables that can plausibly behave as either macroeconomic or financial indicators across regimes. The 17 macroeconomic and 8 financial variables retain their conventional classifications because their economic nature is unambiguous—industrial production is a real-sector series, the overnight interbank rate is a financial-market series—and this classification provides the anchoring structure that pins down the interpretation of $h_{m,t}$ and $h_{f,t}$ as Taiwan's domestic macroeconomic and financial uncertainty, respectively. Without such anchors, the two common factors would be statistically identified but economically interchangeable; with them in place, the loadings of the 22 unclassified variables acquire a direct substantive reading.

To see why $\pi_{i,t}$ functions as a transmission-channel identifier rather than as a conventional regime-switching indicator, consider the U.S. Federal Funds Rate. In a given month, the posterior probability $\pi_{i,t}$ that this variable belongs to the macroeconomic regime takes a value in $[0, 1]$. When $\pi_{i,t} \to 1$, the model's posterior assigns the Federal Funds Rate's common volatility component to $h_{m,t}$: innovations in U.S. monetary policy uncertainty co-move most strongly with Taiwan's real-sector uncertainty, indicating that the transmission occurs primarily through real channels—suppressed export demand, contracted industrial production, delayed investment, and softer manufacturing employment. When $\pi_{i,t} \to 0$, the same shock instead loads onto $h_{f,t}$: its effects operate through financial channels such as capital outflow pressure, NTD/USD depreciation, credit-spread widening, and heightened equity-market volatility. More precisely, because $\pi_{i,t}$ is a time-varying posterior probability rather than a fixed parameter, the same external variable may switch channels across episodes, with the switching dynamics governed by the variable-specific transition probabilities $q^{(i)}_{mm}$ and $q^{(i)}_{ff}$ that are estimated from the data. Time-series plots of $\pi_{i,t}$ for each of the external unclassified variables therefore constitute a direct, data-driven narrative of how transmission mechanisms have evolved over the sample period—particularly across the key episodes of the 2008–2009 Global Financial Crisis, the 2015 China slowdown, the 2018–2019 U.S.–China trade war, the 2020 COVID-19 shock, and the 2022 Federal Reserve tightening cycle. It becomes possible, for the first time in the Taiwan literature, to ask and answer whether a given external shock struck the island through the real side of the economy or through its financial markets, and whether the operative channel in 2022 was the same as the one in 2008.

This reorientation of the unclassified-variables device stands in sharp contrast to its original deployment by Davidson, Hou and Koop (2025). In their application to the U.S. economy, the unclassified block was populated by domestic series whose macroeconomic or financial nature is genuinely ambiguous—S&P 500 returns, the Federal Funds Rate, house prices, and nominal exchange rates—in order to answer the question of which type of uncertainty, macroeconomic or financial, dominates over the U.S. business cycle. The question was inherently domestic, and the unclassified block served as an agnostic holding area for domestic series that might plausibly belong to either of the two anchored blocks. The present project repurposes the same statistical device to answer a fundamentally different, and uniquely small-open-economy, question: through which channel do external shocks enter the domestic economy? In our application, the unclassified block contains all external drivers—four U.S. series, three Chinese series, three global indicators, and the U.S. trade-policy uncertainty measure—while also allowing Taiwan's own ambiguous policy, money, credit, asset-price, housing, exchange-rate, and short-selling variables to be classified by the data rather than by assumption. In contrast to Davidson, Hou and Koop (2025)'s use of the device for domestic classification alone, $h_{m,t}$ and $h_{f,t}$ in our application are anchored by 17 clearly classified macroeconomic variables and 8 clearly classified financial variables, ensuring that the two common factors retain a well-defined economic interpretation as Taiwan's domestic macroeconomic and financial uncertainty. The 22 unclassified variables then load onto these anchored factors in proportions determined entirely by the data, yielding a direct reading of the operative transmission mechanism at each point in time. This reorientation transforms the classification tool of Davidson, Hou and Koop (2025) into a transmission-channel identifier for external shocks—a substantive methodological contribution that goes beyond a country replication exercise and provides a template for applying the OI-SVMVAR framework to any small open economy exposed to competing sources of external uncertainty.

(1.3) Order-Invariance and the Incompatibility of Block Exogeneity

The identification strategy introduced in subsection (1.1)—an unrestricted contemporaneous impact matrix $\mathbf{B}_0$ whose off-diagonal elements are all freely estimated—represents a deliberate departure from the recursive Cholesky scheme that remains the workhorse of large-VAR applications. Under a Cholesky decomposition, $\mathbf{B}_0$ is constrained to be lower triangular, so that variable k may be contemporaneously affected only by variables $1, \ldots, k-1$. This recursive structure imposes an ordering on the elements of $\mathbf{y}_t$ that has no economic justification: for an N-variable system there are $N!$ possible orderings, and in a 47-variable model this amounts to more than 10^{59} admissible permutations, none of which is privileged by economic theory. Davidson, Hou and Koop (2025) document empirically that, in large-scale Cholesky-based SVMVARs, the resulting impulse response functions and forecast error variance decompositions depend substantively on whether a given variable is placed early or late in the ordering, with the same shock producing quantitatively distinct responses across permutations. This arbitrary sensitivity to ordering motivates their order-invariant alternative, whose mechanics—because zero restrictions on $\mathbf{B}_0$ interact with it in a non-obvious way—merit formal exposition before we explain why the standard small open economy (SOE) restriction of block exogeneity cannot be layered on top.

The order-invariant sampler of Davidson, Hou and Koop (2025) resolves the ordering problem through a reparameterization of the rows of $\mathbf{B}_0$. Let $\mathbf{b}_{0,i}$ collect the $N-1$ free off-diagonal elements in the i-th row of $\mathbf{B}_0$, after imposing the normalization $B_{0,ii} = 1$. Conditional on the remaining rows of $\mathbf{B}_0$, the stochastic volatilities, and the VAR residuals, the posterior density of $\mathbf{b}_{0,i}$ is nonstandard because the determinant term $|\det(\mathbf{B}_0)|^T$ enters the likelihood. Davidson, Hou and Koop (2025) therefore introduce an affine transformation of the free row parameters,

$$\mathbf{w}_i = \mathbf{d}_i + \mathbf{Q}_i\, \mathbf{b}_{0,i}, \tag{6}$$

where $\mathbf{d}_i$ and $\mathbf{Q}_i$ are functions of the remaining rows of $\mathbf{B}_0$. This transformation isolates the determinant term in one scalar component of $\mathbf{w}_i$, allowing the sampler to draw that component from an absolute-normal distribution and the remaining components from a conditional Gaussian distribution. Draws of $\mathbf{b}_{0,i}$ are then recovered through the inverse affine mapping. The key point for this proposal is not the algebra of the sampler but its implication: because $\mathbf{B}_0$ is unrestricted apart from the diagonal normalization, the posterior and resulting inference are invariant to permutations of the variables. Impulse responses, variance decompositions, and classification probabilities therefore do not change when the 47 variables are reordered.

This machinery resolves the identification obstacle introduced by a large information set, but it does so under a specific structural assumption—that each row of $\mathbf{B}_0$ is unrestricted apart from the diagonal normalization. Any attempt to superimpose the zero restrictions that underlie block exogeneity changes the posterior problem the DHK sampler was designed to solve. In conventional SOE VARs, researchers routinely partition the variables into an external block (in our context, the U.S., Chinese, and global variables) and a domestic block (Taiwan), and then impose that the external block is permitted to affect the domestic block but not vice versa. Mechanically, this is operationalized as zero restrictions on the rows of $\mathbf{B}_0$ corresponding to external variables: for each external variable i, all elements of $\mathbf{b}_{0,i}$ corresponding to domestic variables are set to zero. If K_i such zero restrictions are imposed on row i, then the sampler no longer draws an unrestricted $(N-1)$-dimensional row vector; it must draw from a lower-dimensional restricted posterior whose normalizing constants, determinant term, and proposal distribution differ from the unrestricted case.

These zero restrictions are therefore not a harmless add-on to Equation (6). The transformation and absolute-normal draw used by Davidson, Hou and Koop (2025) are derived for an unrestricted row of $\mathbf{B}_0$ with one diagonal normalization. Once direction-specific zeros are imposed, generic draws from the unrestricted transformed posterior will violate the zero restrictions, while simply deleting restricted elements changes the determinant term that makes the row posterior nonstandard in the first place. Recovering valid posterior inference under block exogeneity would require a new restriction-aware Gibbs step for $\mathbf{B}_0$, including a different affine transformation, prior support, and proposal distribution. It would also reintroduce a form of ordering or block-design dependence: the result would depend on which variables are designated as external and domestic before estimation, rather than allowing the large information set to discipline the contemporaneous covariance structure. For this project, that would defeat the main purpose of adopting DHK's order-invariant framework.

The appropriate response is not to abandon order invariance, but to replace the zero-restriction approach with the unclassified variables mechanism described in subsection (1.2). By placing all external variables in the unclassified block, this project imposes no zero restrictions on any row of $\mathbf{B}_0$: the full DHK MCMC algorithm is preserved without modification, and the Q-transform in Equation (6) remains valid for every row of the contemporaneous impact matrix. Crucially, this does not mean that the SOE assumption is discarded; rather, it is relocated from a prior restriction to a post-estimation diagnostic. The economic claim that Taiwan's domestic macroeconomic and financial conditions do not systematically drive U.S. monetary policy or Chinese industrial production is assessed ex post via the forecast error variance decomposition described in subsection (2.2): if the share of forecast error variance of each external variable attributable to Taiwan's two domestic uncertainty factors $h_{m,t}$ and $h_{f,t}$ is negligible across all horizons, the SOE property holds approximately in the data. Any sizeable share would indicate that the SOE restriction is counterfactual and ought not to have been imposed in the first place. This design is both methodologically principled and substantively more informative: it delivers a testable SOE assumption—one whose validity emerges from the data—rather than an untested assumption hardwired into the identification scheme.

(2) Anticipated problems and means of resolution

(2.1) Data Assembly and Preprocessing

The empirical strategy set out in subsections (1.1)–(1.3) is demanding of the data: identifying order-invariant posterior classifications for external drivers against anchored domestic factors requires a balanced panel that is simultaneously broad in cross-section, long in time, and harmonized across frequency and vintage. To meet these requirements, we assemble a monthly dataset of $N = 47$ variables spanning **2003M1 through 2025M12**, comprising 276 monthly observations. The dataset is partitioned into the three blocks used by the OI-SVMVAR: 17 macroeconomic variables, 8 financial variables, and 22 unclassified variables, with the complete list and source attribution reported in Table 1 of subsection (1.2). The Taiwan macroeconomic block is drawn primarily from the Directorate-General of Budget, Accounting and Statistics (DGBAS), supplemented by customs trade data from the Ministry of Finance, export-order data from the Ministry of Economic Affairs, and the National Development Council's business-cycle indicator. The Taiwan financial block combines short-rate and bond-yield series from the Central Bank of the Republic of China (CBC) with equity-market, turnover, foreign-investor, and margin-financing series from the Taiwan Stock Exchange (TWSE). TAIEX monthly returns and realized volatility are computed from the TWSE total return index (MFI94U), while the TAIEX log level is retained from the standard price index (MI_5MINS_HIST) because it represents the market price level rather than a reinvested wealth index. The unclassified block draws on CBC monetary and credit aggregates, the Sinyi housing price index, FRED, the BIS effective exchange-rate database, *PolicyUncertainty.com*, the National Bureau of Statistics of China (NBS), the People's Bank of China (PBoC), and the Caldara and Iacoviello (2022) Geopolitical Risk dataset.

The sample begins in 2003M1 for two related reasons. First, the TWSE total return index, which is the cleanest measure of equity-market returns because it incorporates cash dividends, starts in January 2003. Using it as the baseline avoids mechanically understating stock-market returns during ex-dividend months, a concern that is quantitatively visible in Taiwan's July dividend season. Because the first complete monthly return requires a previous month-end total-return observation, the baseline return series begins in 2003M2, while the volatility series is available from 2003M1 using within-month daily observations. Second, beginning in 2003 preserves the post-WTO integration regime while excluding the 1997–1998 Asian financial crisis, a severe structural break in regional financial linkages. The resulting sample contains 276 monthly observations for levels, volatility, and turnover, and 275 complete monthly stock-return observations, which remains large for a monthly Bayesian VAR of this class and covers the key episodes of interest: SARS, the 2008–2009 Global Financial Crisis, the 2015 China slowdown, the 2018–2019 U.S.–China trade war, the 2020 COVID-19 shock, and the 2022 Federal Reserve tightening cycle. To verify that the shorter equity-return series does not drive the findings, a robustness exercise will replace the total return index with the longer TAIEX price index and extend the sample back to 2001M1.

All series are harmonized to a common monthly frequency and transformed to achieve stationarity before estimation. Following Davidson, Hou and Koop (2025), the transformation protocol is applied uniformly to each variable class and proceeds in seven steps. *First*, all level series—including industrial production, employment, retail and food-service sales, exports and imports, price indices, monetary aggregates, and U.S. and Chinese production indices—are expressed as 12-month log-differences (year-on-year log-changes), which removes deterministic seasonality and induces stationarity in a single step while preserving the interpretation of each observation as a growth rate. *Second*, short-term interest rate levels—in particular the overnight interbank rate, the CBC discount rate, and the U.S. Federal Funds Rate—are expressed as monthly first differences, since their near-unit-root behavior would otherwise render the levels non-stationary over the 2003–2025 sample. *Third*, the term spread and credit spreads are retained in levels, because they are already stationary by construction as the difference of two rates; additional differencing would over-whiten these series and eliminate the low-frequency variation that carries the financial-cycle signal. *Fourth*, realized volatility measures for the TAIEX total return index and for the NTD/USD exchange rate are retained in levels, as they are non-negative by construction and, in the present sample, display mean-reverting behavior consistent with stationarity. *Fifth*, seasonal adjustment via X-13-ARIMA-SEATS is applied to any series not already seasonally adjusted by the source agency; DGBAS's seasonally adjusted series are accepted as published, and FRED series apply Census X-13 at source. *Sixth*, the augmented Dickey–Fuller (ADF) and KPSS tests are applied to every transformed series at the 5% significance level, with rejection under the ADF null of a unit root combined with non-rejection under the KPSS null of stationarity taken as joint evidence of stationarity. Any series that fails to attain stationarity under both tests receives an additional first difference and is retested; this safeguard ensures that the balanced panel entering the OI-SVMVAR is stationary in the pre-estimation sense required by the Bayesian algorithm. *Seventh*, all transformed series are standardized to zero mean and unit variance prior to estimation, following Davidson, Hou and Koop (2025), so that the prior distributions on the VAR coefficient matrices $\mathbf{B}_l$ and on the stochastic-volatility-in-mean matrices $\mathbf{A}_j^h$ are scale-invariant across variables with widely different units and levels of variance.

Missing observations arise occasionally for a small number of Chinese and housing-market series in the early months of the sample, where monthly publication practice lagged international conventions or where the source is quarterly. For each affected series, we apply linear interpolation when the gap is at most two consecutive observations; quarterly housing-price observations are converted to monthly frequency using cubic-spline interpolation and checked against lower-frequency movements. For longer gaps, the sample start date for the full panel is trimmed to the earliest month at which all 47 series are observed. The complete record of each raw series, its source identifier, the transformation applied, the stationarity test results, and any interpolation decisions is maintained in a replication log and accompanying codebook, ensuring that the entire data-construction pipeline is reproducible from the original public-source downloads through the balanced panel that enters Equation (1).

(2.2) Works Planned for the First Year

The first year of the project is organized around four overlapping activities that move the empirical framework from raw data to preliminary posterior results. *The first activity*, spanning approximately months 1–4, is the assembly and verification of the dataset. All 47 variables are downloaded from the sources enumerated in Section (2.1), harmonized to a common monthly frequency, and assembled into the balanced panel spanning 2003M1–2025M12. Missing observations—concentrated in a small number of Chinese and housing-market series during the early sample—are addressed via interpolation or, when necessary, by trimming the sample start date. The transformation protocol described in Section (2.1) is applied to each series, and the ADF and KPSS stationarity tests are executed and logged. The complete dataset, the transformation decisions, and the source documentation are recorded in a replication log that accompanies the final panel.

The second activity, spanning approximately months 3–7 and concurrent with the final phase of data assembly, is the implementation and adaptation of the MCMC algorithm. The order-invariant SVMVAR sampler of Davidson, Hou and Koop (2025) is implemented and adapted to the 47-variable Taiwan dataset. The algorithm proceeds as a Gibbs sampler, cycling through the following blocks: the VAR coefficient matrices $\mathbf{B}_l$ and stochastic-volatility-in-mean matrices $\mathbf{A}_j^h$ in Equation (1); the contemporaneous impact matrix $\mathbf{B}_0$, sampled via the row-wise parameter transformation in Equation (6); the common log-volatility paths $h_{m,t}$ and $h_{f,t}$ in Equation (4), drawn in a single block using the band- and sparse-matrix methods of Cross et al. (2023); the idiosyncratic log-volatility paths $\eta_{i,t}^k$; and the Markov-switching classification paths $(s_{i,1}, \ldots, s_{i,T})$ and transition probabilities, drawn using the algorithm of Chib (1996). Convergence is assessed through the spectral diagnostic of Geweke (1992), visual inspection of MCMC trace plots, and effective sample size calculations. Estimation is conducted on the university high-performance computing cluster.

The third activity, spanning approximately months 5–9, is preliminary estimation and benchmarking. Upon completion of the sampler, a preliminary run is conducted on a reduced 30-variable model—retaining the core Taiwan domestic anchors together with a subset of the most important external drivers—to validate the MCMC implementation and to benchmark the estimated $h_{m,t}$ and $h_{f,t}$ paths against the published results of Davidson, Hou and Koop (2025) for the U.S. economy. Upon successful validation, the full 47-variable model is estimated. Each MCMC run requires approximately 30 hours at 50,000 post-burn-in draws, based on the computational requirements reported by Davidson, Hou and Koop (2025) for a model of comparable scale.

The fourth activity, spanning approximately months 9–12, initiates the three-step analysis proper. Posterior draws from the full model are used to extract time-series of the classification probabilities $\pi_{i,t}$ for all 22 unclassified variables, with interpretation centered on the external drivers in the U.S., China, global-indicator, and risk blocks. Preliminary plots are produced for the key episodes identified in Section 1 (the 2008–2009 Global Financial Crisis, the 2015 China slowdown, the 2018–2019 U.S.–China trade war, the 2020 COVID-19 shock, and the 2022 Federal Reserve tightening cycle). A preliminary forecast error variance decomposition is computed to assess the relative contributions of U.S., Chinese, and global uncertainty shocks to Taiwan's domestic factors $h_{m,t}$ and $h_{f,t}$, and preliminary findings are presented at a domestic or international conference in the final quarter of Year 1. Throughout the year, the project research assistant receives structured training in Bayesian econometrics and MCMC methods, building the technical capacity required for the full three-step analysis and for the conditional structural modelling work of Year 2.

1.2 Anticipated results and achievements

The Second Year

1.3 Methods, procedures, and implementation schedule

(1) Research principles, methods, and the innovation of research methods

(1.1) Theoretical Wedge Signatures: The Wedge Projection Theorem

While the Year 1 order-invariant stochastic volatility in mean vector autoregression (OI-SVMVAR) identifies the presence and macroeconomic impact of external uncertainty shocks, it is silent on the underlying micro-founded frictions driving these responses. Conventional approaches attempt to bridge this empirical-to-theoretical gap by matching impulse response functions (IRFs) of standard macroeconomic variables (e.g., output, investment) with those generated by various dynamic stochastic general equilibrium (DSGE) models. However, this variable-level IRF matching is often confounded by general equilibrium effects—such as endogenous interest rate policy and wage adjustments—making it difficult to isolate the exact structural friction.

To overcome this limitation, the second year of this project develops a novel, theorem-based framework: the **Wedge Projection Theorem**. We formalize the exact mathematical mapping between a nonlinear, micro-founded structural model with stochastic volatility (the true data-generating process) and a first-order Business Cycle Accounting (BCA) prototype (the observer's measurement tool).

Specifically, we prove that when the true economy features second-order risk corrections driven by stochastic volatility—denoted as $\Phi(\Omega_t)$ in the nonlinear Euler equation—these corrections mathematically project onto specific first-order wedges in a BCA prototype. For instance, the theorem demonstrates that the BCA investment wedge ($\hat{\tau}_{x,t}$) satisfies the projection equation:

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2}\Phi(\Omega_t), \tag{7}$$

where ω_τ is a steady-state coefficient. This theoretical breakthrough allows us to transition from informally comparing overall IRF shapes to rigorously deriving falsifiable **"wedge signatures."** By solving structural models via third-order perturbation (to capture the independent role of volatility shocks), we can predict precisely *which* wedge an uncertainty shock will perturb and with *what sign*, conditional on a specific friction.

(1.2) Competing Hypotheses: Collateral vs. Working Capital Channels

Armed with the Wedge Projection Theorem, we pre-specify two theoretically motivated candidate mechanisms for the transmission of financial uncertainty, each linked to a distinct, mutually exclusive theoretical wedge signature.

Hypothesis 0: The Collateral and Risk Premium Channel (Second-Order Euler Effect). Following a Fernandez-Villaverde and Guerron-Quintana (FVGQ) style environment, uncertainty tightens borrowing capacity via an occasionally binding collateral constraint. This friction manifests as a second-order risk correction in the intertemporal Euler equation. *Theoretical Signature:* The Wedge Projection Theorem proves that this mechanism projects primarily into the **investment wedge** ($\tau_{x,t}$). Specifically, an increase in financial uncertainty generates a strictly *positive* first-order response in the projected investment wedge ($\kappa_{x,e} > 0$).

Hypothesis 1: The Working Capital Channel (First-Order Static Effect). Alternatively, if firms must finance a fraction of their wage bill in advance, an uncertainty-induced spike in external finance spreads acts as a direct tax on labor demand. This friction operates through a first-order static condition rather than a second-order intertemporal risk correction. *Theoretical Signature:* The theorem proves that this mechanism projects entirely into the **after-tax labor wedge** ($\Lambda_{\ell,t}$). An increase in financing spreads generates a strictly *negative* first-order response in the labor wedge ($\hat{\Lambda}_{\ell,t} < 0$), without requiring third-order Hessian terms.

By deriving these signatures, we establish a targeted diagnostic test: if financial uncertainty predominantly distorts the empirical investment wedge with a positive sign, the data supports the FVGQ collateral channel. If it primarily deteriorates the labor wedge, the evidence favors the working-capital channel.

(2) Anticipated problems and means of resolution

(2.1) Analytical Complexity of Third-Order Perturbation

Deriving the theoretical wedge signatures requires third-order perturbation of the nonlinear DSGE models to isolate the independent effect of stochastic volatility. This introduces substantial analytical and computational complexity, particularly when ensuring that the risk-adjusted steady state is properly defined. To resolve this, we will heavily leverage Dynare's higher-order perturbation routines equipped with pruning to eliminate explosive sample paths. The analytical derivations of the $\Phi(\Omega_t)$ mapping will be cross-validated numerically using Monte Carlo simulations and common-random-number finite difference methods on the policy functions to ensure the theoretical wedge signatures exactly match the numerical DSGE dynamics.

(2.2) Works planned for the second year

The second year focuses exclusively on developing the theoretical framework and formalizing the Wedge Projection Theorem. In the first half of the year, we construct the baseline nonlinear small open economy DSGE models corresponding to the two competing hypotheses: the FVGQ-style collateral model and the wage-bill working capital model. In the second half, we perform the third-order perturbation solutions, derive the analytical mapping between the structural frictions and their respective BCA wedge signatures (κ_x and λ_{wc}), and conduct rigorous numerical validation using Dynare. This step concludes with a formal theoretical paper documenting the Wedge Projection Theorem and the exact diagnostic conditions for distinguishing macroeconomic uncertainty transmission channels.

1.4 Anticipated results and achievements

The Third Year

1.5 Methods, procedures, and implementation schedule

(1) Research principles, methods, and the innovation of research methods

(1.1) Empirical Validation via Business Cycle Accounting

The third year of the project is dedicated to empirical validation, comparing the theoretical wedge signatures derived in Year 2 against Taiwan's actual macroeconomic data. We construct a first-order Business Cycle Accounting (BCA) prototype for a small open economy using Taiwan's aggregate data (e.g., consumption, investment, labor hours, and output). Using standard BCA techniques, we extract the historical time series of the four fundamental empirical wedges: the efficiency wedge, the labor wedge, the investment wedge, and the government/net-export wedge. This procedure strips away general equilibrium dynamics, condensing all complex macroeconomic movements into these four distinct "accounting" components.

(1.2) Targeted Wedge-Level IRF Matching

To determine which structural mechanism dominates in Taiwan, we project the empirical uncertainty factors—specifically, the macroeconomic uncertainty ($h_{m,t}$) and financial uncertainty ($h_{f,t}$) extracted from the SVMVAR in Year 1—onto the empirical wedges extracted in the first step of Year 3. We execute this using local projections (LP) to compute the empirical impulse responses of the individual wedges to uncertainty shocks at horizon $h = 0$ and subsequent periods.

This transforms the standard practice of matching variable-level IRFs into a rigorous **targeted wedge-level IRF matching** exercise. Instead of comparing the ambiguous overall decline in investment, we compare the specific response of the investment wedge and the labor wedge. For example, if the empirical local projection reveals that financial uncertainty ($h_{f,t}$) significantly raises the investment wedge on impact ($\hat{\tau}_{x,0} > 0$), we statistically accept Hypothesis 0 (the collateral/risk premium channel). Conversely, if the shock predominantly deteriorates the labor wedge ($\hat{\Lambda}_{\ell,0} < 0$), we accept Hypothesis 1 (the working capital channel).

(2) Anticipated problems and means of resolution

(2.1) MCMC Uncertainty Propagation and Generated Regressor Bias

A common econometric flaw in "two-step" estimation strategies—where unobserved factors extracted in step one are used as regressors in step two—is generated regressor bias, which occurs when the first-stage estimation uncertainty is ignored.

To resolve this, we implement full **MCMC Uncertainty Propagation**. In our BCA validation step, we will not rely on a single posterior mean path for the uncertainty factors. Instead, we will pass the entire Markov Chain Monte Carlo (MCMC) posterior distribution (e.g., 10,000 retained draws) of the stochastic volatility paths ($h_{m,t}$, $h_{f,t}$) from the Year 1 OI-SVMVAR into the Year 3 wedge local projections. The final confidence intervals for the wedge IRFs will be computed by integrating over the first-stage posterior distribution. This rigorous econometric design ensures that our conclusions regarding the dominant transmission channel are not artifacts of point-estimate approximations, but are robust to the full estimation uncertainty of the underlying shocks.

(2.2) Works planned for the third year

The final year of the project proceeds in three phases. In the first phase, we estimate the baseline BCA prototype for Taiwan and successfully extract the full history of the empirical wedges. In the second phase, we execute the MCMC-integrated local projections of the Year 1 SV factors onto the empirical wedges, generating the empirical wedge-level IRFs. In the third phase, we conduct the formal hypothesis testing by comparing the empirical wedge IRFs against the theoretical signatures derived in Year 2 to conclusively identify the dominant transmission channel for external uncertainty to Taiwan. Finally, based on the verified structural mechanism, we will formulate targeted policy recommendations for the Central Bank of Taiwan (e.g., credit market stabilization versus broad demand stimulus) and finalize the manuscript for journal submission.

1.6 Anticipated results and achievements

- Extraction of empirical Business Cycle Accounting wedges for Taiwan using a small open economy prototype.
- MCMC-integrated local projections of Year 1 uncertainty factors ($h_{m,t}$, $h_{f,t}$) onto empirical wedges, overcoming generated regressor bias.
- Formal statistical identification of the dominant transmission channel (Collateral vs. Working Capital) by testing empirical wedge IRFs against theoretical wedge signatures.
- Targeted policy recommendations for the Central Bank of Taiwan regarding optimal interventions during episodes of external uncertainty, and a completed manuscript for journal submission.

2 Integrated research project

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